Detecting Logic Discrepancies in Cross-Layer Communications within Mini-Program Frameworks

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1 Detections Results on WeChat

Table 1. Zero-day logic discrepancies in cross-layer communications of WeChat.

Issue	Discrepancies			Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O^-	C^-	U^-
C1	GW | EW GS	EW | GW	N/A	⑦	*	*	*	trace monitor	none	button tapping	❶	✔	✘	✔	✔
C2	GW | EW GS	EW | GW	N/A	⑥	boolean	*	wx:if	page monitor	none	button tapping	❶	✔	✔	✔	✔
C3	GW | EW GS	EW | GW	N/A	⑦	boolean	*	wx:if	page monitor	none	button tapping	❶	✔	✔	✔	✔
C4	GS | EW GW	N/A	EW | GS	⑥	*	*	*	trace monitor	none	reloading	❶	✘	✘	✔	✘
C5	GS | EW GW	N/A	EW | GS	⑦	*	*	*	trace monitor	none	reloading	❶	✘	✘	✔	✘
C6	EW | GW GS	EW | GW	EW | GS	③	array	*	content	value monitor	none	button tapping	❸	✘	✔	✔	✔
C7	EW | GW GS	EW | GW	EW | GS	③	array	*	attribute	value monitor	none	button tapping	❸	✘	✔	✔	✔
C8	EW | GW GS	EW | GW	EW | GS	⑦	array	*	attribute	value monitor	none	button tapping	❸	✘	✔	✔	✔
C9	GS | EW GW	N/A	EW | GS	②	function	*	*	trace monitor	none	none	❸	✘	✔	✔	✔
C10	EW | GW GS	EW | GW	EW | GS	①	array	*	wx:for-item	page monitor	conflicting	none	❹	✘	✔	✘	✔
C11	EW | GW GS	EW | GW	EW | GS	②	array	*	wx:for-item	page monitor	conflicting	none	❹	✘	✔	✘	✔
C12	EW | GW GS	EW | GW	EW | GS	③	array	*	wx:for-item	page monitor	conflicting	button tapping	❹	✘	✔	✘	✔
C13	EW | GW GS	EW | GW	EW | GS	⑤	array	*	wx:for-item	page monitor	conflicting	none	❹	✘	✔	✘	✔
C14	EW | GW GS	EW | GW	EW | GS	⑥	array	*	wx:for-item	page monitor	conflicting	none	❹	✘	✔	✘	✔
C15	EW | GW GS	EW | GW	EW | GS	⑦	array	*	wx:for-item	page monitor	conflicting	button tapping	❹	✘	✔	✘	✔
C16	GS | EW GW	N/A	EW | GS	⑦	*	*	*	trace monitor	coexisting (data)	button tapping	❺	✔	✘	✘	✔
C17	GS | EW GW	N/A	EW | GS	⑦	*	*	*	trace monitor	coexisting (element)	button tapping	❺	✘	✘	✘	✔
C18	GS | EW GW	N/A	EW | GS	③ ⑥	*	*	*	trace monitor	coexisting (all)	button tapping	❺	✘	✘	✘	✔
C19	GS | EW GW	N/A	EW | GS	④ ⑥	*	input	*	trace monitor	coexisting (all)	text entering	❺	✘	✘	✘	✘
C20	GS | EW GW	N/A	EW | GS	⑥ ⑦	*	*	*	trace monitor	coexisting (all)	button tapping	❺	✘	✘	✘	✔

Issue	Discrepancies			Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O^-	C^-	U^-
	EW	GW	GS												
P1			N/A	⑥	*	*	content	trace monitor	none	none	❷	✘	✘	✔	✔
P2			N/A	⑦	*	*	content	trace monitor	none	none	❷	✘	✘	✔	✔
P3	N/A			①	*	*	*	page monitor	none	none	❸	✔	✔	✔	✔
P4	N/A			⑤	*	*	*	page monitor	none	none	❸	✔	✔	✔	✔
P5	N/A	N/A		⑨	*	*	attribute	value monitor	none	button tapping	❸	✔	✘	✔	✘
P6			N/A	③ ⑥	*	*	*	trace monitor	coexisting (all)	none	❺	✔	✘	✘	✔
P7			N/A	③ ⑦	*	*	*	trace monitor	coexisting (all)	none	❺	✔	✘	✘	✔
P8			N/A	⑦	*	*	*	trace monitor	none	button tapping	❻	✘	✘	✔	✔

Table 1 reports the detection results of MiniDiff on WeChat respectively(for paper §6.3).

In the `Discrepancies` column, the icon means platform: Android (), iOS (), Windows (), macOS (), and DevTools (DEV) for WeChat. The letter means configuration: Exparser+WebView ($\mathbb{EW}$), GlassEasel+Skyline ($\mathbb{GS}$), GlassEasel+WebView ($\mathbb{GW}$).

In the `Data Channel` column, the number (①-⑨) means the types of Data Channel: Channel ① (`RJS` $\overset{var}{\rightsquigarrow}$ `XML`), Channel ② (`RJS` $\overset{fun}{\rightsquigarrow}$ `XML`), Channel ③ (`RJS` $\overset{upd}{\rightsquigarrow}$ `XML`), Channel ④ (`RJS` $\overset{upd}{\leftrightsquigarrow}$ `XML`), Channel ⑤ (`EJS` $\overset{var}{\rightsquigarrow}$ `XML`), Channel ⑥ (`EJS` $\overset{fun}{\rightsquigarrow}$ `XML`), Channel ⑦ (`RJS` $\overset{var}{\rightsquigarrow}$ `EJS` $\overset{fun}{\rightsquigarrow}$ `XML`), Channel ⑧ (`XML` $\overset{ev}{\rightsquigarrow}$ `RJS`), Channel ⑨ (`XML` $\overset{ev}{\leftrightsquigarrow}$ `EJS`).

In the `Root Cause` column, the number(❶-❻) means the types of root causes: ❶ inconsistent shadow-data maintenance, ❷ inconsistent life-cycle management, ❸ inconsistent object (de)serialization, ❹ confused messaging channel, ❺ inconsistent virtual DOM diffing, and ❻ inconsistent thread model.

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Table 2. Zero-day logic discrepancies in cross-layer communications of Alipay.

Issue	Discrepancies	Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O^-	C^-	U^-
C21	$\mathbb{WR}$ | $\mathbb{NR}$	⑥	*	*	*	trace monitor	none	reloading	❶	✘	✘	✔	✘
C22	$\mathbb{WR}$ | $\mathbb{NR}$	⑦	*	*	*	trace monitor	none	reloading	❶	✘	✘	✔	✘
C23	$\mathbb{WR}$ | $\mathbb{NR}$	①	function	*	content	value monitor	none	none	❺	✘	✔	✔	✔
C24	$\mathbb{WR}$ | $\mathbb{NR}$	②	function	*	*	value monitor	none	none	❺	✘	✔	✔	✔
C25	$\mathbb{WR}$ | $\mathbb{NR}$	③	function	*	content	value monitor	none	button tapping	❸	✘	✔	✔	✔
C26	$\mathbb{WR}$ | $\mathbb{NR}$	⑤	function	*	content	value monitor	none	none	❸	✘	✔	✔	✔
C27	$\mathbb{WR}$ | $\mathbb{NR}$	⑨	*	*	attribute	value monitor	none	button tapping	❺	✘	✘	✔	✘

Issue	Discrepancies	Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O^-	C^-	U^-
P9	DEV |	①	array	*	attribute	page monitor	none	none	❸	✘	✔	✔	✔
P10	DEV |	③	array	*	attribute	page monitor	none	button tapping	❸	✘	✔	✔	✔
P11	DEV |	⑧	*	*	*	value monitor	none	button tapping	❺	✔	✘	✔	✘

2 Detections Results on AliPay

Table 2 reports the detection results of MiniDiff on AliPay respectively(for paper §6.3).

In the `Discrepancies` column, the icon means platform: Android (), iOS () and DevTools (DEV) for AliPay

The letter means configuration: WebView rendering ($\mathbb{WR}$) and native rendering ($\mathbb{NR}$).

In the `Data Channel` column, the number (①-⑨) means the types of Data Channel: Channel ① (RJS $\overset{var}{\rightsquigarrow}$ XML), Channel ② (RJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ③ (RJS $\overset{upd}{\rightsquigarrow}$ XML), Channel ④ (RJS $\overset{upd}{\leftrightsquigarrow}$ XML), Channel ⑤ (EJS $\overset{var}{\rightsquigarrow}$ XML), Channel ⑥ (EJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ⑦ (RJS $\overset{var}{\rightsquigarrow}$ EJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ⑧ (XML $\overset{ev}{\rightsquigarrow}$ RJS), Channel ⑨ (XML $\overset{ev}{\leftrightsquigarrow}$ EJS).

In the `Root Cause` column, the number(❶-❻) means the types of root causes: ❶ inconsistent shadow-data maintenance, ❷ inconsistent life-cycle management, ❸ inconsistent object (de)serialization, ❹ confused messaging channel, ❺ inconsistent virtual DOM diffing, and ❻ inconsistent thread model.

Table 3. Zero-day logic discrepancies in cross-layer communications of TikTok.

Issue	Discrepancies	Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O^-	C^-	U^-
	$\mathbb{WR}$												
P12	DEV, Android, iOS	⑥	*	*	content	trace monitor	none	none	❷	✘	✘	✔	✔
P13	DEV, Android, iOS	⑦	*	*	content	trace monitor	none	none	❷	✘	✘	✔	✔
P14	DEV, Android, iOS	①	array	*	a:for	page monitor	none	none	❸	✔	✔	✔	✔
P15	DEV, Android, iOS	③	array	*	a:for	page monitor	none	none	❸	✔	✔	✔	✔
P16	DEV, Android, iOS	⑤	array	*	a:for	page monitor	none	none	❸	✔	✔	✔	✔
P17	DEV, Android, iOS	③ ⑥	*	*	*	trace monitor	coexisting (all)	none	❺	✘	✘	✘	✔
P18	DEV, Android, iOS	③ ⑦	*	*	*	trace monitor	coexisting (all)	none	❺	✘	✘	✘	✔

3 Detections Results on TikTok

Table 3 reports the detection results of MiniDiff on TikTok respectively(for paper §6.3).

In the `Discrepancies` column, the icon means platform: Android (), iOS () and DevTools (DEV) for TikTok. The letter means configuration: WebView rendering ($\mathbb{WR}$).

In the `Data Channel` column, the number (①-⑨) means the types of Data Channel: Channel ① (RJS $\overset{var}{\rightsquigarrow}$ XML), Channel ② (RJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ③ (RJS $\overset{upd}{\rightsquigarrow}$ XML), Channel ④ (RJS $\overset{upd}{\leftrightsquigarrow}$ XML), Channel ⑤ (EJS $\overset{var}{\rightsquigarrow}$ XML), Channel ⑥ (EJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ⑦ (RJS $\overset{var}{\rightsquigarrow}$ EJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ⑧ (XML $\overset{ev}{\rightsquigarrow}$ RJS), Channel ⑨ (XML $\overset{ev}{\leftrightsquigarrow}$ EJS).

In the `Root Cause` column, the number(❶-❻) means the types of root causes: ❶ inconsistent shadow-data maintenance, ❷ inconsistent life-cycle management, ❸ inconsistent object (de)serialization, ❹ confused messaging channel, ❺ inconsistent virtual DOM diffing, and ❻ inconsistent thread model.